

LESSON 5.3

POINTS ON THE COORDINATE PLANE – PART 1



LESSON INTRODUCTION

MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x- or y-axis as the line of reflection when two ordered pairs have an opposite x- and y-coordinate.

LEARNING TARGETS

- I can plot rational numbers in the form of fractional ordered pairs.
- I can plot rational numbers in the form of decimals.

BENCHMARK COHERENCE

BUILDING ON

MA.5.GR.4.1 Identify the origin and axes in the coordinate system. Plot and label ordered pairs in the first quadrant of the coordinate plane.

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How can you determine the scale for ordered pairs that include decimals?
- How can you determine the scale for ordered pairs that include fractions?
- How can you use the least common multiple to determine the scale for ordered pairs that include fractions?

LESSON NARRATIVE

In the previous lesson, students plotted points with rational values with different scales. This lesson builds on previous understanding to further explore graphing coordinates with rational numbers. The 5.3.1 Warm-Up is an opportunity for students to reflect on the previous lesson of choosing the correct scale, a better scale than what is listed. Students will expand their practice of choosing a correct scale with decimal units and fractional units and work collaboratively to plot fraction and decimal numbers on the coordinate plane.

COMPONENT SUMMARY

5.3.1 WARM-UP (~5 MIN.)

Choose the best graph to represent a data set

5.3.2 COLLABORATION (10 MIN.)

Plot decimals on a coordinate plane

5.3.3 TRY IT (5-10 MIN.)

Create a scale to plot decimal points on the coordinate plane

5.3.4 GUIDED INSTRUCTION (15-20 MIN.)

Plot points on a fractional scale

5.3.5 TRY IT (5 MIN.)

Plot points on a fractional scale independently

5.3.6 WRAP-UP (~5 MIN.)

Reflect on understanding of learning targets

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Scale
- Least Common Multiple

MATERIALS

- (Optional) Dry erase boards
- (Optional) Dry erase markers

ADDITIONAL PREPARATION

N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not understand how to find the least common denominator.
- Students may struggle with linking the scale to the least common denominator.

THINGS TO CONSIDER

- Students are still grasping rational numbers as well as negative numbers. Consider reviewing what these numbers look like in a visual context to help understanding.
- If time is an issue, consider assigning 5.3.5 Try It for homework.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

5.3.4 Guided Instruction provides an opportunity for students to engage in a Think-Pair-Share routine to formalize their explanations with a partner before engaging with the written-response questions.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

This lesson encourages students to work collaboratively with their partners to reason about plotting points on a coordinate plane. Encourage students to analyze their partner's explanations and critique or build upon responses as necessary. Through partner work, students engage in mathematical reasoning and have the opportunity to justify and explain their work.

DIFFERENTIATE INSTRUCTION

Highlight and make space for the differentiated approaches to sensemaking, which are embedded throughout this lesson.

- For verbal processors, or students who learn best when thinking and reasoning with others, the 5.3.2 Collaboration will drive their learning and offer an opportunity for making meaning of fluency with scales.
- For students who learn best with trial and error or seeing the work in action, consider providing a graphing utility as they work through the 5.3.2 Collaboration.

5.3.1 WARM-UP: WHICH ONE IS THE BEST?

Component Overview: In this Warm-Up, students choose a graph to best represent the data presented and explain why their choice is appropriate.

ENRICHMENT

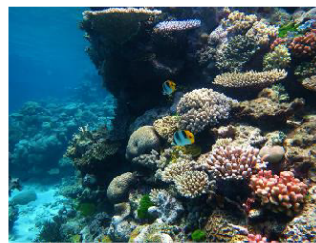
This Warm-Up positions students to reason about the appropriateness of scales on coordinate graphs. To help students begin to make connections between the data points and choosing an appropriate scale, ask questions like:

- What do you notice about the temperatures in the table?
- What scale would most clearly show these values? Why?
- How can you use the data points in a set to help determine an appropriate scale?



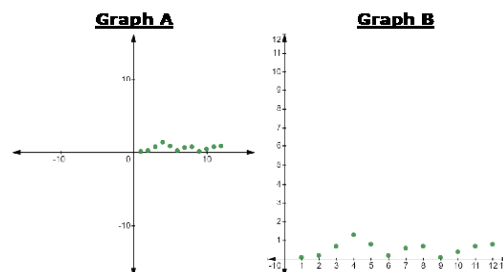
5.3.1 Warm-Up: Which One is the Best?

Coral polyps get stressed out and could die if the surrounding water is above 29°C . The table represents the average temperature, to the nearest tenth, above 29°C at the Great Coral Reef over a 12-week period.

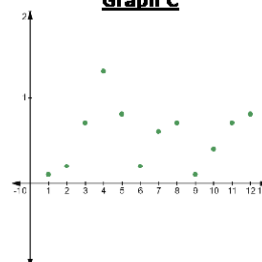


1. Explain which graph best represents the data. (Data taken from NOAA.)

Week	1	2	3	4	5	6	7	8	9	10	11	12
Temp	0.1	0.2	0.7	1.3	0.8	0.2	0.6	0.7	0.1	0.4	0.7	0.8



Graph C

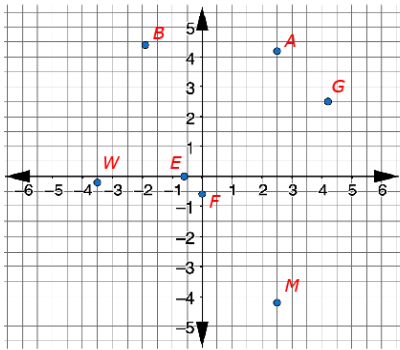


Answers will vary. Graph C best represents the data because the x-axis accurately represents all the integers used in the data for weeks. The scale for the y-axis depicts the smallest and largest data point to show the variation in the data.



5.3.2 Collaboration: Plotting Points with a Decimal Scale

1. A coordinate plane is shown.



a. Working with your partner, describe the scale that is being used for each axis.

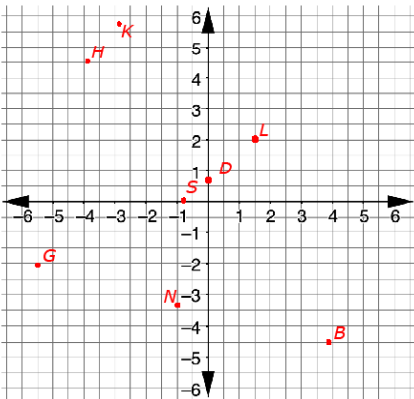
Answers will vary. The scale is 0.5 or $\frac{1}{2}$ because it takes 2 lines to get to each whole number. 1 whole divided by 2 is 0.5 or $\frac{1}{2}$.

b. Locate each point on the coordinate plane and label it with the appropriate letter.

Point G (4.2, 2.5)	Point B (-1.9, 4.4)	Point F (0, -0.6)	Point W (-3.5, -0.2)
Point E (-0.6, 0)	Point A (2.5, 4.2)	Point M (2.5, -4.2)	

2. Working with your partner, plot and label the points using the given letters on the grid.

Point L (1.5, 2)	Point B (3.9, -4.5)	Point G (-5.5, -2.1)	Point D (0, 0.7)
Point K (-2.8, 5.7)	Point S (-0.8, 0)	Point N (-1, -3.4)	Point H (-3.9, 4.5)



a. Which points were easy to plot?

Answers will vary. Point L, Point D, Point H and Point S

b. Which points were more difficult to plot?

Answers will vary. Point B, Point G, Point N, Point K

5.3.2 COLLABORATION: PLOTTING POINTS WITH A DECIMAL SCALE

Component Overview: In this Collaboration, students will work with a partner to plot decimals on a coordinate plane. Students will practice plotting points on a graph that uses whole numbers for the scale and discuss why or why not using this type of graph may work.

ENRICHMENT

To encourage students to begin to draw conclusions about the scale on the graph, consider asking guiding questions like:

- When graphing Point B, should the x-coordinate be closer to -1 or -2?
- What do you notice about where each unit on the scale falls? How many lines apart are they? What does this tell you about the scale?
- What do you know about points that don't fall directly on the lines where the numbers are?

DIFFERENTIATE INSTRUCTION

Consider allowing students to graph the points using a graphing utility if they are struggling to plot the points on the graph.

TEACHER MOVES

Specifically, have students make connections to which points were easy to plot and which were more difficult. Encourage students to reason about what scales might be appropriate for decimal numbers and fractions, making note that points that were not exactly halfway between two numbers might have been more difficult to plot.

5.3.2 COLLABORATION: PLOTTING POINTS WITH A DECIMAL SCALE (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider providing students with sentence starters to help scaffold their thinking on their own and when speaking with their partner.

- “To determine the best scale to use when plotting decimal values, you can...”
- “I know how to choose which scale to use by considering...”

3. On your own, answer the following question:

When plotting ordered pairs with decimal values, how is it best to determine the scale to use when setting up the axes on a grid?

My Reasoning

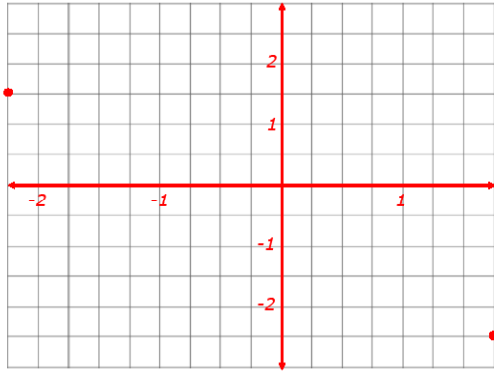
Answers will vary. To determine the scale to use when plotting ordered pairs with decimals determine any common factors in the coordinates. For example, if all of the x-coordinates have one decimal place and the tenths place is 0 or 5, then the scale I would choose would be .5 or $\frac{1}{2}$.

Share your reasoning with your partner and ask for feedback on any parts that were unclear or missing. Make any necessary edits to your reasoning.



5.3.3 Try It: Plotting Points with a Decimal Scale

- When plotting $(1.75, -2.5)$ and $(-2.25, 1.5)$, describe how to best scale the axes.
Answers will vary. 0.25 or $\frac{1}{4}$ would be the best scale for the x-axis because when looking at the coordinates the decimal values are all multiples of 0.25. 0.5 or $\frac{1}{2}$ would be the best scale for the y-axis because both y-coordinates are multiples of 0.5.
- For the points $(1.75, -2.5)$ and $(-2.25, 1.5)$, draw and label coordinate axes with an appropriate scale, and plot the points.



5.3.3 TRY IT: PLOTTING POINTS WITH A DECIMAL SCALE

Component Overview: In this Try It, students will be given a blank graph and will need to create a decimal scale and plot decimal points accordingly. Students will need to look at each ordered pair to determine the best pattern of numbers.

DIFFERENTIATE INSTRUCTION

Consider allowing students to plot the points on a dry erase board before writing in their workbooks. This will allow students to use trial and error to arrive at an answer, as they can erase a whiteboard more easily.

5.3.4 GUIDED INSTRUCTION: PLOTTING POINTS WITH A FRACTIONAL SCALE

Component Overview: In this Guided Instruction, students will practice plotting points that are fractions on a coordinate plane. Students will need to create a fractional scale to accurately plot each point. Students will also draw and create their own graph using fractional points. This activity is also available via Desmos.

ENRICHMENT

Students may struggle to plot the fractional points on the graph. Encourage students to reason about the appropriate scale by asking questions like:

- How many fractional parts should the space between 0 and 1 be broken up into to accurately plot each point?
- Is there another fractional scale that can be used for this graph?
- Does the scale need to be the same for both the x- and y-axes?

DIFFERENTIATE INSTRUCTION

Some students may still struggle with x- and y-coordinates and plotting them accurately on the graph. Encourage students to create a table to visually see the x- and y-values of each coordinate point to help them plot correctly on the graph.

TEACHER MOVES

Notice the use of the term “better” in relation to which scale is being used. Facilitate a brief discussion on what it means to be better in terms of fluency. Is there one correct answer? Could both work? What does that have to do with fluency?



5.3.4 Guided Instruction: Plotting Points with a Fractional Scale

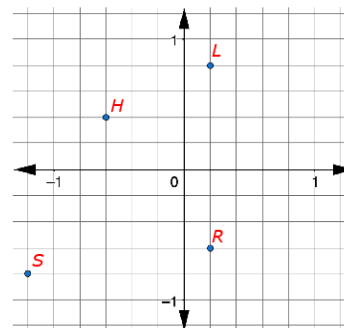
1. Four points are shown.

Point L	Point H	Point S	Point R
$(\frac{1}{5}, \frac{4}{5})$	$(-\frac{3}{5}, \frac{2}{5})$	$(-1\frac{1}{5}, -\frac{4}{5})$	$(\frac{1}{5}, -\frac{3}{5})$

- a. Based on the denominators of each fraction, what scale should be used for the grid?

$\frac{1}{5}$

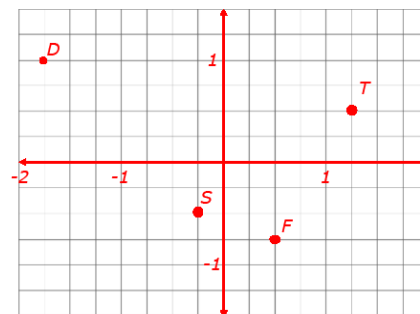
- b. Find and label the points with the given letters on the coordinate plane.



2. Four different points are shown.

Point F	Point T	Point S	Point D
$(\frac{1}{2}, -\frac{3}{4})$	$(1\frac{1}{4}, \frac{1}{2})$	$(-\frac{1}{4}, -\frac{1}{2})$	$(-1\frac{3}{4}, 1)$

- a. List the denominators of each fraction.
2 and 4
- b. Liana says when she sets up her grid, she will use a scale of $\frac{1}{4}$. Trey says he plans to use a scale of $\frac{1}{2}$. Explain who is planning to use the better scale.
Answers will vary. Liana is using the better scale because the fractions be written with a denominator of 4.
- c. Draw and label coordinate axes with the scale of your choice. Then, plot and label the points.



3. Another set of four points is shown.

Point M	Point L	Point C	Point A
$(\frac{2}{3}, \frac{1}{6})$	$(\frac{1}{2}, -\frac{1}{3})$	$(-\frac{5}{6}, -\frac{1}{2})$	$(-\frac{1}{6}, 1)$

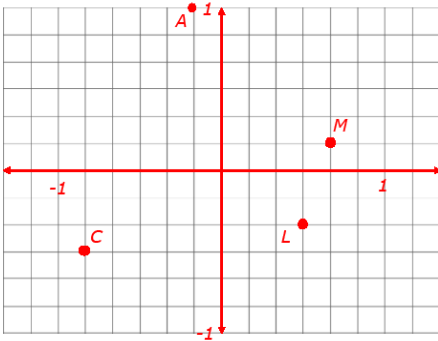
- a. List the denominators of each fraction.
2, 3, 6

- b. Zoe says she does not want to label her axes by halves because she will only be able to approximate the location of the thirds. She also says she does not want to label by thirds because there is no easy way to approximate the location of the halves. Christopher says she should label her axes by sixths.

Explain the benefit of using Christopher's plan.

Answers will vary. Christopher's plan will allow him to plot fractions with denominators of 2 and 3 because he will be able to write those as equivalent fractions with a denominator of 6.

- c. Draw and label coordinate axes with the scale you chose and plot the points.



When deciding on a scale for plotting ordered pairs with rational values, consider the least common multiple (LCM) of each of the denominators to determine the proper increments.

5.3.4 GUIDED INSTRUCTION: PLOTTING POINTS WITH A FRACTIONAL SCALE (CONTINUED)

TEACHER MOVES

Think-Pair-Share

Engage students in a Think-Pair-Share routine as they discuss and complete question 3b. This will give students time to analyze each student's response and determine why Christopher's plan is beneficial. Then students discuss with a partner to encourage mathematical discourse and reasoning.

5.3.5 TRY IT: PLOTTING POINTS WITH FRACTIONS AS A SCALE

Component Overview: In this Try It, students will practice plotting points with a fractional scale. Students will need to decide which fractional amount to use for the graphing by finding a common denominator.

ENRICHMENT

Students may struggle to determine an appropriate scale because the fractions have different denominators. Further scaffold this section through questioning like:

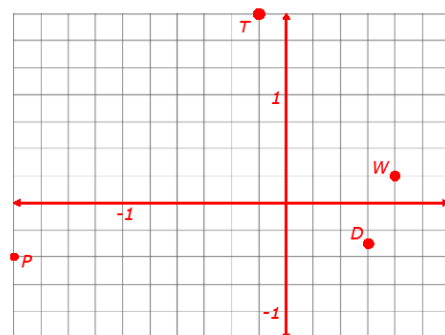
- What is a common denominator that can be used for all of these coordinates? What's the least common multiple, and how can you find it?
- How can you convert the fractions so they all have the same denominator?
- Do the x- and y-axes need to have the same scale? Why or why not?



5.3.5 Try It: Plotting Points with Fractions as a Scale

1. Draw and label coordinate axes with an appropriate scale. Then, plot and label the points.

Point W	Point D	Point P	Point T
$(\frac{2}{3}, \frac{1}{4})$	$(\frac{1}{2}, -\frac{1}{3})$	$(-\frac{5}{2}, -\frac{1}{2})$	$(-\frac{1}{6}, \frac{2}{4})$

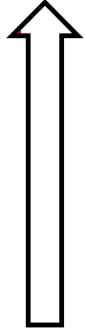




5.3.6 Wrap-Up: Reflection

1. Shade the arrow up to the statement that best describes your current level of understanding.

Answers will vary.



I'm so confident, I could explain plotting ordered pairs made up of fractions and/or decimals to someone else!

I can get to the right answer, but I don't understand it well enough to explain it yet.

I understand some of this, but I don't understand all of it yet. I checked the boxes I think I do not understand.

I tried hard and I listened, but I am finding this challenging. I checked the boxes I think I do not understand.

2. Check the skills that you currently understand.

Answers will vary.

- ☐ Determining the correct scale
- ☐ Identifying the LCM of the denominators for coordinates that include fractions
- ☐ Plotting ordered pairs that contain fractions
- ☐ Plotting ordered pairs that contain decimals

5.3.6 WRAP-UP: REFLECTION

Component Overview: In this Wrap-Up, students will reflect on their understanding of the lesson's learning targets. Consider using this Wrap-Up as a guide for which students will need reteaching or reinforcing of the lessons the next day.

DIFFERENTIATE INSTRUCTION

Consider providing additional support for struggling readers by reading the checklist of skills out loud or showing examples of each skill in context.